Edward Silva

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Experience

Software Engineering Intern, Kratos Defense — Jun – Aug 2025
C++, Python, SIMD, Git — Colorado Springs, CO

- Sped up legacy real-time DSP code in C++ by 1.6x through refactoring, AVX-512 vectorization, and Intel VTune profiling on Linux over SSH
- Presented a faster FIR filter implementation and its performance trade-offs to the team for use on later projects
- Built a configurable logging framework with severity levels and error tracking to cut debug time and improve maintainability of the real-time DSP software

Projects

Non-Linear State Estimation for the EMBR Damper, MATLAB, Simulink — Spring 2026

- Designed a non-linear state estimation strategy for the semi-active EMBR suspension, modeling the interactions of a custom magnetorheological (MR) fluid damper, MOSFET power regulation, and a multi-sensor array
- Derived continuous and discrete-time non-linear state-space models in MATLAB and Simulink, simulating system kinematics to validate damping performance against harmonic, stochastic, and discrete terrain disturbances
- Implemented and compared Extended (EKF) and Unscented (UKF) Kalman Filters to estimate unmeasured states

Autonomous Maze-Navigating Robot, Python — Fall 2025

- Achieved the second-fastest course completion time in the cohort with an autonomous maze-navigating robot built by a 4-person team, integrating computer vision, embedded control, and I2C communication
- Built the computer vision subsystem with one teammate, using Python and OpenCV with ArUco marker detection to extract real-time distance and angle metrics for navigation
- Designed a multithreaded vision pipeline processing 640x480 video feeds with HSV color masking to identify directional markers for pathfinding logic
- Interfaced the vision subsystem with motor control via I2C bus, transmitting telemetry data to an Arduino while updating a real-time RGB LCD display

Skills

Skills: C++, Python, C, SIMD, MATLAB, Simulink, Git, Arduino

Education

Colorado School of Mines — May 2026
BS Electrical Engineering, Minor in Computer Science
Relevant Coursework: Signals and Systems I/II, Algorithms, Digital Signal Processing, Estimation Theory & Kalman Filtering, Modern Control Design, Embedded Systems, Computer Architecture, Software Engineering